PLANTFORGE: A Procedural Control-Plant Corpus for In-Context System Identification

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Abstract

In-context system-identification (SysID) transformers are typically trained on a single private data-generation recipe — most commonly Wiener-Hammerstein-only multisine (or similarly narrowband) data, regenerated independently by every paper in this line of work. We introduce PLANTFORGE, a released, static, reproducible corpus of synthetic control-plant trajectories, generated on the fly from a procedural distribution (not sampled from the static released shards, which are a separate, versioned snapshot for reproducibility and identifiability-annotation reference; Section 3) and organized along three axes jointly — nonlinearity *family*, excitation *class*, and sampling *rate* (exact zero-order-hold) — with per-instance Fisher-information identifiability annotations. Across 5 independently-seeded training runs, we show that a transformer trained on Wiener-Hammerstein-only data collapses off its training distribution (cross-family nMSE degradation of 112–179$\times$), and that training on PLANTFORGE's multi-axis corpus closes the rate and excitation gaps (to $\sim$1–2$\times$ for four of five families; the resonant drivetrain family improves but stays at 8.8$\times$) while substantially narrowing, but not closing, the cross-family gap (a $\sim$6$\times$ reduction in degradation, 112–179$\times$ $\rightarrow$ 17–31$\times$). We then subject this result to three checks a released benchmark should survive but rarely does: (1) zero-shot transfer to three real measured plants (Silverbox, Cascaded Tanks, Bouc-Wen), where corpus-trained models again outperform the Wiener-Hammerstein-only model, but a simple classical ARX baseline outperforms *both* trained transformers by 6–284$\times$ on every one of the three, tempering any claim of transformer superiority; (2) a within-cell statistical re-analysis showing that the corpus's own identifiability annotations do *not* positively predict in-context prediction difficulty (median Spearman $r = -0.112$ across 40 cells), the opposite of our working hypothesis; and (3) full disclosure of a Simpson's-paradox confound our first analysis of (2) missed, and how we caught and corrected it. We release the corpus (240,000 instances, CC BY 4.0), all generation/training/evaluation code (MIT), and every number in this paper with reproduction instructions.

1 Introduction

Model-agnostic, in-context system identification — training a sequence model to infer a plant's dynamics from a short context of (u, y) pairs, with no gradient updates at test time — is an increasingly popular paradigm for control and mechatronics [Forgione et al., 2023, Piga et al., 2024]. The training recipe behind this line of work, however, is narrow and largely unreleased: each paper privately regenerates Wiener-Hammerstein-only multisine (or similarly narrowband) data, the one family the incumbent generators support. No existing corpus varies *nonlinearity family*, *excitation class*, and

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sampling rate jointly, and none ships per-instance identifiability annotations that would let a user reason about *why* an instance is hard.

This paper makes two kinds of contribution. First, we release PLANTFORGE: a procedurally generated corpus spanning 5 nonlinearity families $\times$ 4 excitation classes $\times$ 3 sampling rates (240,000 instances total), with per-instance relative Cramér-Rao lower bounds (CRLB) and Fisher-information matrix (FIM) condition numbers as identifiability annotations, and design invariants enforced by tests (Section 3). Second, and we think more valuably for the community, we report what happened when we subjected our own headline claim to increasingly rigorous scrutiny: multi-seed error bars, zero-shot real-plant validation, a classical baseline, and a statistical re-analysis of our identifiability hypothesis. Two of these checks produced results that *complicate* the paper's original thesis rather than confirm it, and we report them in full rather than adjusting the narrative around them (Section 4).

Summary of findings.

- Multi-axis training on PLANTFORGE closes the rate and excitation transfer gaps that Wiener-Hammerstein-only training leaves open (for four of five families; the resonant drivetrain family improves but does not close), but only substantially narrows, not closes, the harder cross-family gap — reported here as an open challenge, not a solved problem (Section 4.1).
- Zero-shot on three real plants, a corpus-trained model beats a Wiener-Hammerstein-only model on every one, but a simple per-window ARX fit beats *both* trained transformers by 6–284$\times$ under an identical, leakage-free evaluation protocol (Section 4.3).
- The corpus's own identifiability annotations, hypothesized to predict in-context prediction difficulty, show a weak, robustly *negative* within-cell correlation with it once a between-cell confound in our first analysis is corrected (Section 4.4).

2 Related Work

Synthetic foundation models for dynamics. Ziegler et al. [2024] pretrain a transformer on synthetic dynamics sampled from a reproducing kernel Hilbert space and show it transfers to real hardware after fine-tuning. This is generically capable but does not vary family/excitation/rate/identifiability jointly, nor target control-plant nonlinearities specifically; we view it as the closest prior synthetic-data approach, and the one whose scope PLANTFORGE is most easily confused with. A separate, theoretical line of work [Cole et al., 2025] derives approximation/depth-separation bounds for in-context learning of *linear* dynamical systems; PLANTFORGE addresses the empirical/benchmark side this theory does not cover.

In-context system identification. The training recipe we stress-test — Wiener-Hammerstein-only multisine pretraining — is used across the Forgione et al. [2023], Piga et al. [2024] line of work (the "forgi86 lineage"), each regenerating its own private WH-only data. Rufolo et al. [2025b] propose an enhanced architecture (probabilistic framing, non-contiguous windows, recurrent patching) for the same task — PLANTFORGE evaluates a plain encoder-only baseline instead, so our findings are not confounded by architecture novelty; Busetto et al. [2024] extend the same paradigm to state estimation, showing the narrow-recipe pattern recurs beyond the one paper we benchmark against; Rufolo et al. [2025a] attack the same generalization gap via a distributionally-robust training *objective*, complementary to our strategy of broadening the training *corpus*. Akram and Vikalo [2024] give an independent theoretical account of what in-context transformers compute for dynamical systems, evidence the paradigm has uptake beyond this one lineage. Our own `InContextSysID` is a standard encoder-only transformer over (u, y) pairs — our contribution is the corpus and its scrutiny, not architectural novelty.

Domain randomization and procedural benchmark generation. PLANTFORGE's design — generate broadly across a parametrized axis so a held-out real system looks like "just another variation" — is the system-identification instance of domain randomization, established in robotics by Tobin et al. [2017] and surveyed by Muratore et al. [2022]; unlike most of that literature, PLANTFORGE directly measures the residual zero-shot gap on real plants (Section 4.2) rather than assuming randomization closes it, and finds it does not fully close for the hardest axis. Procedural generation as a benchmark-design tool in its own right is established by Cobbe et al. [2020] for reinforcement learning and by Takamoto et al. [2022], Gupta and Brandstetter [2022] for PDE surrogate modeling (Section 3's three jointly-varied axes are PLANTFORGE's analog of these platforms' varied PDE type/parameters/boundary conditions, plus an excitation-design axis these PDE benchmarks have no equivalent of). The general phenomenon PLANTFORGE's transfer-gap measurement instantiates is studied at ML-wide scale by Koh et al. [2021].

Scaling benchmarks and other released dynamics corpora. DynaDojo [Bhamidipaty et al., 2023] benchmarks how SysID algorithms scale with training samples, system complexity, and target error over $\sim$20 generic ODE/chaotic systems, but without control-specific nonlinearities (deadzone, saturation, hysteresis, drivetrain compliance), an excitation/rate axis, or identifiability annotations. Ullah and Baca [2026] release a real-hardware (Crazyflie 2.1) multi-task SysID/control/state-estimation benchmark, evidence the community values multi-axis design even outside the synthetic setting PLANTFORGE occupies.

Real-measured benchmarks. The community's default real-data benchmarks — Silverbox, Wiener-Hammerstein, Bouc-Wen [Noël and Schoukens, 2020], Cascaded Tanks [Wigren and Schoukens, 2013, Schoukens and Ljung, 2019] — are fixed, single-record systems with no parameter ground truth. We evaluate zero-shot against three of four (Section 4.2); a classical baseline outperforming our trained transformers on them (Section 4.3) is itself evidence a synthetic corpus with known parameters is a useful complement, not replacement, for this suite.

Classical nonlinear system identification and identifiability. PLANTFORGE's ARX/NARX baseline (Section 4.3) and per-instance Fisher-information identifiability annotations (Section 3, tested in Section 4.4) sit within a mature classical literature: black-box nonlinear model structures are surveyed by Sjöberg et al. [1995], and NARMAX methods — including the free-run instability our degree-2 NARX baseline reproduces — by Billings [2013]. Our Fisher-information/Cramér-Rao formalism is standard theory [Ljung, 1999]; Hjalmarsson [2005] surveys experiment/excitation design's link to identification accuracy, against whose usual assumption (better excitations $\rightarrow$ better identifiability $\rightarrow$ more useful data) our null result in Section 4.4 is a direct counterpoint. Aguirre et al. [2010] study the same prediction-error-vs.-simulation-error distinction our ARX protocol turns on; excitation design specifically — multisine and PRBS, both in our `excitation.py` — is treated at length by Pintelon and Schoukens [2012].

3 The PLANTFORGE Corpus

Design axes. Every instance is generated along three axes jointly: **family** — `stribeck` (velocity-dependent friction, a state-nonlinearity), `backlash` (input deadzone), `saturate` (input clipping), `boucwen` (output hysteresis with memory), and `drivetrain` (two-inertia motor/gear/compliant-load with load Coulomb friction, the mechatronics staple) — plus a Wiener-Hammerstein (`wh`) family excluded from the corpus proper but retained for the failure experiment in Section 4.1; **excitation** — `prbs` (0.25 s physical hold), `multisine` (8 tones $\leq$1.85 Hz), `chirp` (0.05$\rightarrow$3 Hz), and `closedloop` (a true sequential PI-tracking loop against the plant stepper, so the same seed against a different plant instance produces different u — not a two-pass imitation of closed-loop data); and **rate** — 10/20/50 Hz, exact zero-order-hold discretizations of one shared continuous-time truth per instance.

Identifiability annotations. Each (instance, excitation, rate) combination carries a per-parameter relative Cramér-Rao lower bound and a Fisher-information-matrix $\log_{10}$ condition number, computed analytically from the same simulator that generated the trajectory.

Design invariants (tested). (1) (u, y) pairs are physically consistent with their stated parameters — no hidden output normalization; re-simulation from θ reproduces y to 10^{-6}. (2) Multi-rate instances are the *same* continuous-time plant: state-nonlinear families substep internally at $\leq$2 ms with zero-order-hold input, so $\mathrm{d}t$ and $\mathrm{d}t/4$ agree at common sample instants to within 10^{-3} relative error (exact for LTI cores) — an axis we are not aware of any other released corpus generating exactly. (3) Closed-loop excitation is a genuine sequential loop, verified by checking that u depends on the specific plant instance under control. (4) Identifiability annotations are directionally physical: an excitation that never exits the `backlash` deadzone yields a relative CRLB on the deadzone width $\sim 5\times 10^4$, versus $\sim$0.3 for an excitation that does.

Release. 240,000 instances (5 families × 4 excitations × 3 rates × 4,000 instances/cell), 583 MB, released under CC BY 4.0 [Nguyen, 2026]. All generation, training, and evaluation code is released under MIT. Full datasheet, following the Gebru et al. [2021] template (motivation, composition, collection process, uses, distribution, maintenance), accompanies the release. This static, versioned snapshot exists for reproducibility, identifiability-annotation reference, and this paper's own Section 4.4 experiment — it is *not* the transformer's literal training set: the `corpus`-trained models in Section 4.1 are trained on fresh draws from the same procedural generator distribution (regenerated per training batch, never read from the released shards), so the released corpus and the transformer's training distribution are related but distinct artifacts.

4 Experiments

All models are an identical encoder-only transformer (`InContextSysID`: 5 layers, 8 heads, width 160) consuming 192 context samples and predicting 32 query-horizon samples with no ground-truth y visible in the query horizon. Every headline number in this section is the mean ± standard deviation over **5 independently-seeded training runs** (seeds 0–4, 10,000 steps each, fixed evaluation seeds 900–905 across every model for paired comparison). nMSE is always the ratio of batch-mean query-horizon squared error to batch-mean query-horizon signal power, never a mean of per-window ratios.

4.1 Headline: multi-axis training closes the rate/excitation gap, narrows the family gap

We compare two training recipes: **wh_only**, trained exclusively on Wiener-Hammerstein multisine data (the incumbent recipe), and **corpus**, trained on 4-of-5 PLANTFORGE families × {`multisine`, `prbs`} × {10, 50 Hz}, with `backlash`, `chirp`/`closedloop`, and 20 Hz held out — in both cases the training batches are drawn fresh from the generator distribution, never from the released static shards (Section 3).

Table 1: Synthetic transfer matrix, nMSE (mean±std, $n = 5$ seeds). ×ref is relative to that model's own in-distribution reference cell.

held-out axis	wh_only nMSE	(×ref)	corpus nMSE	(×ref)
reference (train-like)	0.0034 ± 0.0001	1.0×	0.0192 ± 0.0007	1.0×
family `backlash`, dt=0.10	0.3800 ± 0.0081	111.6×	$\mathbf{0.3205 \pm 0.0147}$	**16.7**×
family `backlash`, dt=0.05	0.3883 ± 0.0085	114.0×	$\mathbf{0.3448 \pm 0.0161}$	**17.9**×
family `backlash`, dt=0.02	0.6112 ± 0.0065	179.4×	$\mathbf{0.5988 \pm 0.0155}$	**31.1**×
excitation `chirp`	0.1417 ± 0.0393	41.6×	$\mathbf{0.0355 \pm 0.0129}$	**1.8**×
excitation `closedloop`	0.0636 ± 0.0162	18.7×	$\mathbf{0.0273 \pm 0.0112}$	**1.4**×
rate dt=0.05, `stribeck`	0.0300 ± 0.0012	8.8×	$\mathbf{0.0210 \pm 0.0017}$	**1.1**×
rate dt=0.05, `saturate`	$\mathbf{0.0140 \pm 0.0005}$	**4.1**×	0.0287 ± 0.0029	1.5×
rate dt=0.05, `boucwen`	0.0869 ± 0.0068	25.5×	$\mathbf{0.0186 \pm 0.0015}$	**1.0**×
rate dt=0.05, `drivetrain`	0.2036 ± 0.0022	59.8×	$\mathbf{0.1691 \pm 0.0108}$	**8.8**×

Table 1 and Figure 1 summarize the result. Multi-axis training **solves the rate and excitation axes for most families**: the WH-only model's 4.1–42× degradation on interpolated rate and held-out excitation collapses to $\sim$1–2× under corpus training for four of the five trained families. The exception is the resonant `drivetrain` family, whose rate cell improves 59.8× → 8.8× but does not close — a second open finding, invisible to single-family generators. Multi-axis training also **substantially narrows, but does not close, the cross-family gap** (a $\sim$6× reduction in the degradation ratio, 112–179× → 17–31×, still an order of magnitude off reference; the reduction is far more modest in absolute nMSE terms, only 11% at dt=0.05 — the ratio and absolute framings tell different-sized stories, and we report both rather than the more-flattering one) — we report this as PLANTFORGE's open challenge track, not a solved problem. This 17–31× figure is the `backlash` holdout specifically; a full leave-one-family-out sweep (Section 5, Table 4) shows it is not a typical family-transfer number — three of the five families show essentially no cross-family gap when held out individually. One further cell, `saturate` at the held-out rate, is won by `wh_only` (non-overlapping error bars the other direction): “corpus training helps” is not a universal law across every cell, and we report the exception rather than omit it. Non-overlapping ± 1 standard deviation across 11 simultaneous cell comparisons is a descriptive heuristic, not a multiplicity-corrected test; at $n = 5$ seeds this is a conservative bar relative to the standard error of the mean, and the largest-margin cells (below) clear it comfortably.

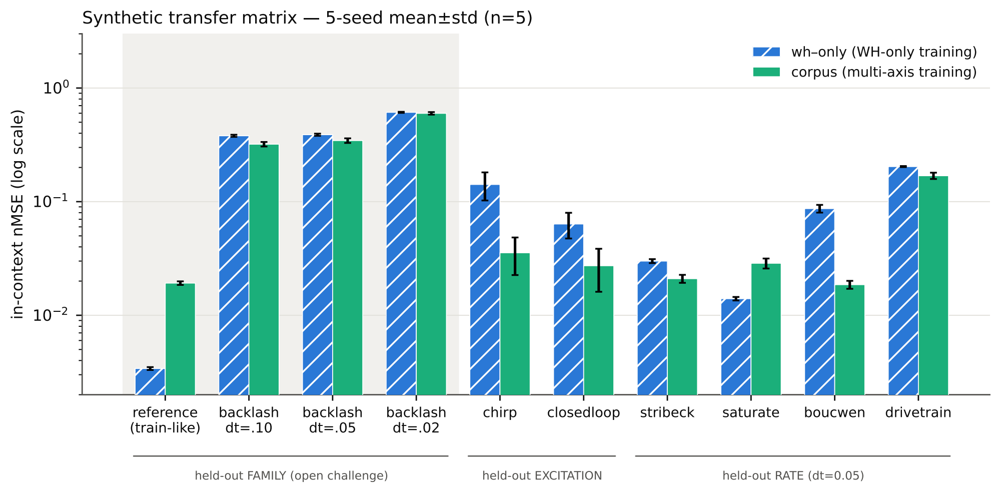

Figure 1: Synthetic transfer matrix, `wh_only` vs. `corpus`, 5-seed mean$\pm$std, log scale.

4.2 Zero-shot transfer to real measured plants

Both checkpoints, trained only on synthetic PLANTFORGE data, are evaluated zero-shot (no fine-tuning) on real recordings from nonlinearbenchmark.org [Wigren and Schoukens, 2013, Schoukens and Ljung, 2019], obtained via the `nonlinear_benchmarks` package for two of the three datasets below, and via a direct fetch from its official source for the third (4TU.ResearchData [Noël and Schoukens, 2020], CC BY-SA 4.0, since Bouc-Wen is not exposed by `nonlinear_benchmarks`’s public loader API). Real sample rates do not align with PLANTFORGE’s three trained rates, so we handle each dataset on its own terms: **Silverbox** (native $\mathrm{d}t \approx$1.64 ms) is decimated 12$\times$ (anti-aliased) to $\mathrm{d}t \approx$19.7 ms, landing near the finest trained rate; **Cascaded Tanks** (native $\mathrm{d}t =$4.0 s) is already 40$\times$ coarser than the coarsest trained rate and is evaluated at its native rate as an explicit extrapolation; **Bouc-Wen** (native $\mathrm{d}t = 1/750$ s exactly) is decimated 15$\times$ to $\mathrm{d}t = 0.02$ s *exactly* (no rounding error), the cleanest rate match of the three — note this is a real hysteretic mass-spring-damper benchmark, distinct from PLANTFORGE’s own synthetic `boucwen` corpus family (Section 3), which models output hysteresis but is not fit to this specific real system; **Wiener-Hammerstein Process Noise** is excluded — its $\sim$1.5 s total record duration is shorter than one 224-sample context window at any trained rate, so it cannot supply a single in-context evaluation window.

Table 2: Zero-shot real-plant nMSE. wh_only/corpus: mean$\pm$std, 5 seeds, fixed windows. ARX/NARX2: single deterministic run, $\leq$8 non-overlapping windows (Section 4.3).

dataset	wh_only	corpus	ARX	NARX2
Silverbox (decimated 12$\times$)	0.7941 ± 0.2098	0.4863 ± 0.0937	**0.0028**	0.0028
Cascaded Tanks (native rate, extrapolation)	0.3653 ± 0.1564	0.0415 ± 0.0121	**0.0073**	3.8×10^9
Bouc-Wen (decimated 15$\times$, exact)	1.9684 ± 0.4163	1.3497 ± 0.1756	**0.0616**	0.0549

Corpus training again beats WH-only training zero-shot on real data (most dramatically on Silverbox), consistent with Section 4.1. This held-up conclusion is itself a worked example of why we report every real-plant number at 5-seed scale: an earlier single-seed (seed 0 only) pass on Bouc-Wen showed the *opposite* direction (corpus nMSE 2.05 vs. wh_only 1.66) — a result two independent reviewers confirmed was not a code defect given the data available at the time, since `wh_only`’s seed-to-seed std on Bouc-Wen (0.4163) is the largest of any real-plant standard deviation in this paper. At full 5-seed scale the direction reverses back to the Silverbox/Cascaded-Tanks pattern. We surface this not to bury it, but because it is exactly the failure mode multi-seed reporting exists to catch — and the reason every number in this paper, including the architecture ablation in Section 5, is now reported at the same 5-seed scale rather than a cheaper single-seed pass.

But Table 2 previews a result that complicates the story further: a trivial classical baseline beats *both* trained transformers by two orders of magnitude on every one of the three real-plant benchmarks. We discuss why in Section 4.3.

4.3 Classical baselines temper the transformer story

We compare against ARX and degree-2 polynomial NARX, fit by ordinary/ridge least squares *per window*, under an evaluation protocol identical to the transformer's: fit (or, for the transformer, condition) on the 192-sample context only, then free-run the 32-sample query given only u — neither method ever observes query-horizon ground-truth y. ARX order $k \in \{2, 4, 8\}$ is selected per window by one-step-ahead error on the last 32 context samples, never touching the query.

Result. Table 2 and Figure 2 show ARX beating both trained transformers on all three real-plant benchmarks by 6–284$\times$ (Bouc-Wen: 22–32$\times$, within the same range). This pattern recurs synthetically: ARX also beats both models on held-out excitation `chirp` (0.0038 vs. 0.0355/0.1417), held-out excitation `closedloop` (0.0058 vs. 0.0273/0.0636), and held-out rate `drivetrain` (0.0254 vs. 0.1691/0.2036), and roughly ties `corpus` on held-out family `backlash` at the coarsest rate (0.3106 vs. 0.3205).

Is this a fair comparison? We verified it is. ARX and the transformer consume identical windows, identical normalization, and an identical context/query split; the transformer's own query-horizon input is zeroed exactly where ARX is forbidden from using ground truth, so neither method sees query y at fit/inference time. ARX's per-window least-squares fit is not an unfair privilege — *adapting to the context* is precisely the in-context transformer's own value proposition, executed via a frozen forward pass instead of an explicit fit. The real advantage ARX has on rate-shifted and real-plant cells is structural, not a protocol bug: a per-window linear refit at the test sampling rate is immune to a rate mismatch that a frozen transformer must instead extrapolate through, most severely on Cascaded Tanks' 40$\times$ extrapolation. We view this as a genuine limitation of the frozen in-context approach worth further study, not an artifact to explain away.

Degree-2 NARX diverges broadly in free-run (clipped-but-huge nMSE, 10^9–10^{12}, on all but two of the twelve evaluated cells) — a legitimate finding about naive polynomial NARX instability under 32-step free simulation, and partly attributable to the $k = 8$ order candidate being numerically underdetermined at the order-selection stage (153 quadratic features vs. 152 fit samples). The two exceptions, both real-plant, are Silverbox and — newly — Bouc-Wen (NARX2 0.0529, close to ARX's 0.0603); we do not have a full account of why free-run quadratic NARX stays stable on these two real systems and not elsewhere, and flag it as an open question rather than force an explanation. We report all NARX2 numbers rather than omit them, but do not treat the diverged ones as comparable to ARX's.

4.4 Identifiability annotations do not predict prediction difficulty

PLANTFORGE's per-instance rel-CRLB and FIM-condition annotations were hypothesized, at design time, to predict in-context prediction difficulty — intuitively, a poorly-identifiable instance should also be hard to predict. We tested this directly: per-instance query-horizon nMSE, scored on the `corpus` model (5 seeds, mean), against each instance's identifiability annotation, on the 40 (family $\times$ excitation $\times$ rate) cells whose trajectory length exceeds one context window (10 Hz cells are excluded: 128 samples $<$ 224).

A confound we caught ourselves. Our first analysis pooled all 40,000 instances and correlated directly: Spearman $r = -0.082$ (max rel-CRLB) and $r = -0.318$ ($\log_{10}$ FIM condition), both statistically significant but with per-family breakdowns of *heterogeneous sign and magnitude* (`backlash` $+0.320$, clearly positive, against a near-zero-to-negative spread for the other four: `boucwen` $+0.006$, `stribeck`-0.126, `saturate`-0.136, `drivetrain`-0.159). This heterogeneity is the fingerprint of a Simpson's-paradox confound: excitation and family choice change both identifiability and model

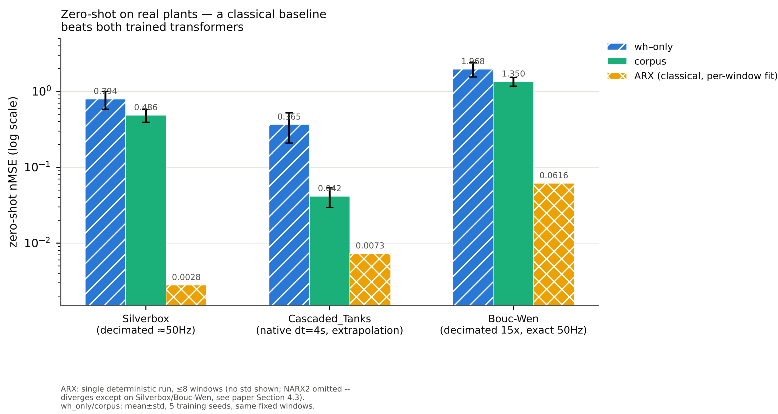

Figure 2: Zero-shot real-plant nMSE: `wh_only`, `corpus`, and ARX. Log scale.

difficulty independently, so a pooled correlation conflates between-cell structure with the within-cell effect the hypothesis is actually about. We discarded the pooled analysis as the primary result and re-ran within-cell.

Corrected analysis. We compute Spearman r within each of the 40 cells separately, then aggregate. We further apply two confound controls: (a) excluding each cell's bottom decile of query-horizon signal power (near-flat query horizons mechanically inflate nMSE, a live confound independent of identifiability), and (b) restricting to cells with non-degenerate identifiability dynamic range (all 40/40 cells pass this filter).

Table 3: Within-cell Spearman r(max rel-CRLB, per-instance nMSE), aggregated over 40 cells, 5 seeds, 1,000 instances/cell (40,000/40,000 finite).

	median r	mean r	cells with $r > 0$
base	-0.112	-0.093	10/40
power-controlled (excl. bottom-decile power)	-0.113	-0.102	—
range-filtered (IQR$>10^{-6}$, 40/40 cells kept)	-0.112	—	—

Result: weak, robustly negative, not the hypothesized direction. The within-cell correlation is small and negative, and survives both confound controls without flipping sign (Table 3). Only 10/40 cells are positive, concentrated in two families: `backlash` (4/8 cells, family mean $+0.017$, driven by its `multisine` and `closedloop` excitations) and `drivetrain` (6/8 cells, family mean -0.008, close to null since its one negative cell, `chirp` at dt=0.05, is a large outlier); r reaches $+0.340$ at its highest (`backlash`, `closedloop`, dt=0.02). `stribeck`, `saturate`, and `boucwen` are negative on balance (family means -0.12 to -0.19, Figure 3).

Interpretation. We propose the negative sign reflects real mechanistic decoupling, not a null result: rel-CRLB measures *parameter-recovery* difficulty, not *prediction* difficulty — a hard-to-identify parameter typically has little influence on the output, so an in-context predictor need not infer what barely affects y. The annotations remain valuable corpus metadata (e.g. for excitation-design studies) but should not be marketed as a prediction-difficulty predictor. A secondary methodological finding: naive quartile-binned *means* of per-instance nMSE show a dramatic apparent upward trend (2.4 $\rightarrow 9.3 \times 10^8$ across quartiles) purely because per-instance nMSE has an extremely heavy right tail; the quartile *medians* are flat (0.035, 0.040, 0.043, 0.021, Figure 4) — our own initial mean-based framing for this experiment was itself a symptom of the same heavy-tail artifact, flagged here so the failure mode is documented, not just quietly fixed.

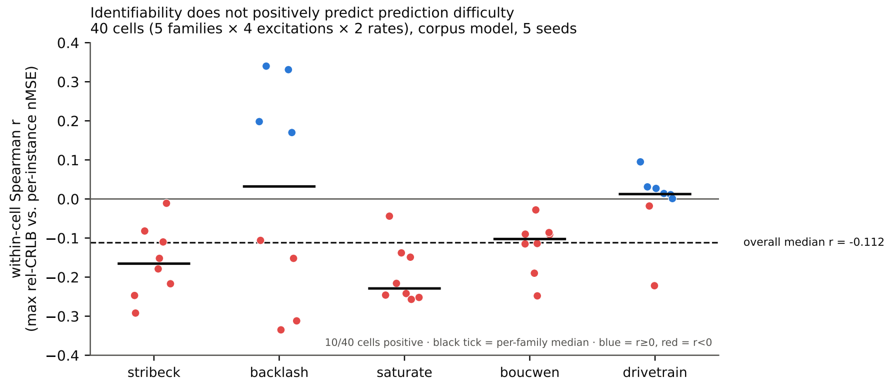

Figure 3: Within-cell Spearman r, all 40 cells, corpus model, 5 seeds. Ticks mark per-family medians.

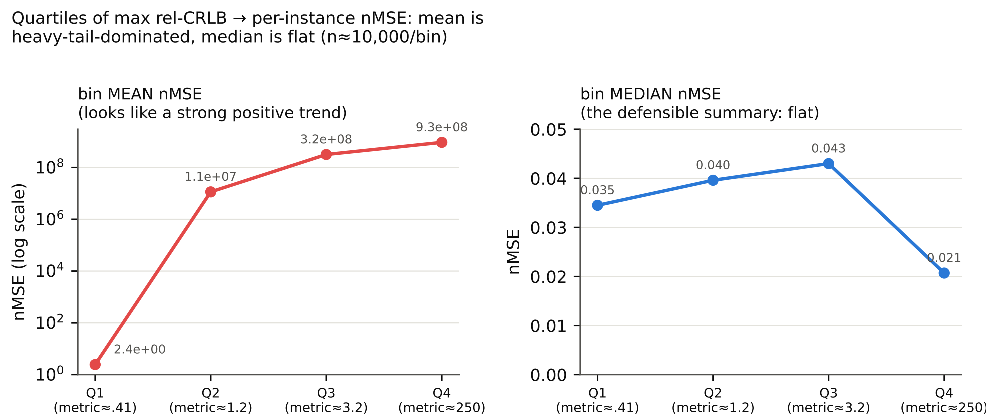

Figure 4: Quartiles of max rel-CRLB $\rightarrow$ per-instance nMSE: bin mean (left, heavy-tail-dominated) vs. bin median (right, flat).

5 Discussion and Limitations

What this paper does and does not claim. We do *not* claim in-context transformers are the strongest available method for real-plant system identification: Section 4.3's ARX result argues against that claim directly. We *do* claim that (a) training-distribution breadth matters — multi-axis training measurably narrows, though does not close, the transfer gaps a narrow WH-only recipe leaves open, and (b) a released, parameter-annotated, three-axis corpus is a more useful research artifact for studying *why* than any of the alternatives in Section 2, independent of whether the transformer architecture itself is currently the best tool for the job.

Scope of real-plant validation. We validate zero-shot on 3 of the 4 real benchmarks named in our design motivation (Silverbox, Cascaded Tanks, Bouc-Wen); only Wiener-Hammerstein Process Noise is excluded, by record length (Section 4.2). Real-plant nMSE uses $\leq$8 non-overlapping windows per dataset (variance is over training seeds, not window sampling); the ARX comparison is a single deterministic run. The 6–284$\times$ margins dwarf plausible window-sampling noise, but this is a first zero-shot probe, not an exhaustive evaluation. Bouc-Wen's `wh_only` variance (std 0.4163 on a mean of 1.9684) is our largest real-plant standard deviation, and a single-seed preview of this cell showed the opposite ranking before the full 5-seed run (Section 4.2) — a reminder that per-cell variance, not just the headline mean, matters.

Family coverage and held-out-family choice. We hold out a single family (`backlash`) throughout Sections 4.1–4.4. A full leave-one-family-out sweep (5 seeds/family, `corpus` recipe) shows the corpus's five families split into two difficulty tiers rather than one representative-or-outlier case

(Table 4). `backlash` and `drivetrain` are indistinguishable at ∼0.33–0.34 held-out nMSE (means differ by less than either's seed-to-seed std, $\pm 1\sigma$ bands overlap) — 5–8× higher than the other three families (∼0.042–0.060), which score *below* the trained-family-average reference when held out (an upper-bound reference averaged over the 4 trained families' two trained rates, including the 2 hard ones, not each easy family's true in-distribution nMSE — read as "nearly free to hold out," not "beats training on it"). This recontextualizes the headline cross-family gap (Section 4.1): the 17–31× figure is specific to `backlash`, not a uniform property of family transfer. `drivetrain`'s difficulty is consistent with it already being hardest for held-out-*rate* transfer (8.8×, Table 1): its resonant two-inertia dynamics resist both transfer axes, while `backlash`'s deadzone is the only other family that does. The corpus's five families still omit multi-input/multi-output plants and time-varying parameters within a trajectory.

Table 4: Leave-one-family-out sweep (mean±std, `corpus` recipe, $n = 5$ seeds every row): reference (mean nMSE over the 4 trained families, averaged over the two trained rates) vs. held-out family nMSE (at the held-out rate dt=0.05), ranked by held-out nMSE.

held-out family	reference nMSE	held-out nMSE	n
backlash	0.0484 ± 0.0029	0.3448 ± 0.0161	5
drivetrain	0.0790 ± 0.0049	0.3277 ± 0.0247	5
saturate	0.1175 ± 0.0046	0.0599 ± 0.0064	5
boucwen	0.1155 ± 0.0035	0.0452 ± 0.0037	5
stribeck	0.1207 ± 0.0114	0.0423 ± 0.0023	5

Architecture ablation: the family gap persists across capacity, and does not resolve cleanly with scale (5-seed check). All headline results (Sections 4.1–4.4) use one fixed transformer configuration (160 width, 5 layers, 1.58M params). We ran a lighter-weight check — `corpus` recipe only, but at the same 5-seed scale as every other number in this paper — across four additional variants spanning a 15× parameter range, to test whether the held-out-family gap is an artifact of this specific size.

Table 5: Architecture ablation (mean±std, 5 seeds, corpus recipe): reference vs. held-out family `backlash` dt=0.05 nMSE.

variant	params	reference nMSE	family nMSE (dt=0.05)
narrow (d=80)	407k	0.0300 ± 0.0028	0.3708 ± 0.0108
default (d=160, L=5)	1.58M	0.0192 ± 0.0007	0.3448 ± 0.0161
shallow (L=2)	655k	0.0419 ± 0.0029	0.3419 ± 0.0094
deep (L=8)	2.51M	0.0155 ± 0.0011	0.3528 ± 0.0088
wide (d=320)	6.24M	0.0170 ± 0.0016	0.3261 ± 0.0172

Along the **width** axis (L=5 fixed), absolute family-transfer nMSE *does* move monotonically with capacity — narrow (0.3708) > default (0.3448) > wide (0.3261) — and the two extremes are genuinely distinguishable, not just noise ($\pm 1\sigma$ bands $[0.3600, 0.3816]$ for narrow and $[0.3089, 0.3433]$ for wide do not overlap): going wider measurably helps. Along the **depth** axis (d=160 fixed), the direction reverses — shallow (0.3419) < default (0.3448) < deep (0.3528), a monotonic *worsening* with depth — but here the extremes do overlap at $\pm 1\sigma$ ($[0.3325, 0.3513]$ vs. $[0.3440, 0.3616]$), so this direction is not distinguishable from noise at $n = 5$. In-distribution reference nMSE, by contrast, improves monotonically with capacity on *both* axes, exactly as expected. Net effect: capacity scaling is not the fix either — even `wide`, the best-performing variant across a 15× parameter range, remains 19.2× its own reference, still an order of magnitude off, and depth scaling shows no benefit at all (if anything, a non-significant cost). We revise our original framing accordingly: the cross-family gap is not simply "capacity-invariant" (width does move it), but it is not a capacity problem in the sense of "add more parameters and it resolves" either — at no point in this 15× range does either axis bring the family gap within reach of the in-distribution reference. The same ×ref ratio trap flagged above (Table 4's discussion) and in Section 4.4 recurs here: `deep`'s worse-looking relative gap (22.8× vs. `shallow`'s 8.2×) is driven substantially by a 2.7× difference in reference nMSE (0.0155 vs. 0.0419), not solely by the (overlapping, not significant) absolute family-transfer difference (0.3528 vs. 0.3419). This check does not cover context/query-length variation; a sweep over those axes is left to future work.

On negative and inconvenient results. Two of this paper's four experiments (Sections 4.3 and 4.4) produced results that complicate rather than support the paper's motivating narrative; we report both in full, including the Simpson's-paradox confound our first identifiability analysis missed and how it was corrected, because an honestly-reported null result and a caught methodological error are more useful to the community than a quieter framing. A third result, the leave-one-family-out sweep (Section 5, Table 4), is likewise inconvenient: our single held-out family was not representative, and the headline cross-family gap is specific to two of five families, not general. We surface it in the same spirit.

6 Broader Impact

PLANTFORGE is a synthetic corpus of control-plant dynamics; we are not aware of misuse vectors beyond those generic to open ML tooling. Positive impact: broadening the training distribution for in-context SysID may reduce duplicated private data-generation effort (the "demand proof" motivating this corpus, Section 2), and the released identifiability annotations support excitation-design research relevant to real experimental efficiency in control engineering.

7 Conclusion

We released PLANTFORGE, a three-axis synthetic control-plant corpus with identifiability annotations, and subjected our central claim (broader training data narrows distribution-shift gaps for in-context SysID transformers) to progressively harder scrutiny: multi-seed error bars, zero-shot real-plant transfer, a classical ARX baseline, a corrected identifiability re-analysis, a 5-seed ablation, and a leave-one-family-out sweep. The claim largely holds up: the `backlash` held-out-family gap persists across a $15\times$ capacity range without resolving — width scaling helps modestly and measurably, depth scaling does not, and even the best-performing variant remains an order of magnitude off reference — and cross-family difficulty concentrates in two of five families (`backlash`, `drivetrain`), not uniformly. Two auxiliary claims did not hold: ARX beats both transformers on every real-plant benchmark and several synthetic cells, and identifiability annotations do not positively predict difficulty. A near-miss on Bouc-Wen (Section 5) shows even our multi-seed standard is not immune to high-variance cells. We report all of this as part of the release.

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8 NeurIPS Paper Checklist

1. **Claims**
   - Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?
   - Answer: [Yes]
   - Justification: The abstract states five concrete, hedged claims: (i) WH-only training collapses off-distribution (cross-family nMSE degradation of 112–179×); (ii) PLANTFORGE's multi-axis corpus closes the rate/excitation gap for 4 of 5 families but only substantially narrows, not closes, the cross-family gap (112–179× → 17–31×); (iii) a classical ARX baseline outperforms both trained transformers on all three real-plant benchmarks by 6–284×; (iv) identifiability annotations do not positively predict prediction difficulty (median Spearman $r = -0.112$ across 40 cells); (v) full disclosure of a Simpson's-paradox confound in our own first identifiability analysis. Each claim is backed by a corresponding experiment (Sections 4.1–4.4) and hedged by a matching Limitations paragraph (Section 5) — e.g., the cross-family-gap claim explicitly notes it is specific to the `backlash` holdout, not a general property of family transfer (Section 4.1, Section 5, Table 4).
2. **Limitations**
   - Question: Does the paper discuss the limitations of the work performed by the authors?
   - Answer: [Yes]
   - Justification: Section 5 devotes six paragraphs to limitations: what the paper does and does not claim; the scope of real-plant validation (window-count and single-ARX-run caveats, 3 of 4 candidate real benchmarks); family coverage and the held-out-family choice (resolved via a 5-seed leave-one-family-out sweep, Table 4); the scope of the architecture ablation (no context/query-length sweep); and an explicit "On negative and inconvenient results" paragraph naming which findings complicate the paper's own motivating narrative.
3. **Theory Assumptions and Proofs**
   - Question: For each theoretical result, does the paper provide the full set of assumptions and a complete and correct proof?
   - Answer: [NA]
   - Justification: The paper contains no theoretical results or proofs; its contribution is an empirical corpus, benchmark suite, and set of controlled experiments.
4. **Experimental Result Reproducibility**
   - Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?
   - Answer: [Yes]
   - Justification: All corpus-generation code (family/excitation/rate definitions, exact zero-order-hold simulation, Fisher-information annotation computation), training code, and evaluation code are released under MIT (Section 3). Every headline number is reported as mean±std over 5 independently-seeded training runs (seeds 0–4) evaluated on a fixed set of evaluation seeds (900–905) shared across all models for paired comparison (Section 4). Training checkpoints are resumable and the release's training scripts are idempotent against already-finished checkpoints.
5. **Open Access to Data and Code**
   - Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results?
   - Answer: [Yes]
   - Justification: The corpus (240,000 instances, CC BY 4.0) is hosted at `https://huggingface.co/datasets/stark4062/plantforge`; all generation, training, and evaluation code (MIT) is hosted at `https://github.com/Soarr01/plantforge`, including a README with exact commands for every experiment reported in this paper.

6. **Experimental Setting/Details**
   - Question: Does the paper specify all the training and test details needed to understand the results?
   - Answer: [Yes]
   - Justification: Section 4's preamble specifies the architecture (5-layer, 8-head, width-160 encoder-only transformer), context/query lengths (192/32 samples), the training-seed count (5, seeds 0–4), and the fixed evaluation-seed range (900–905). Section 4.3 specifies the ARX/NARX2 order-selection protocol ($k \in \{2, 4, 8\}$, selected by one-step-ahead error on the last 32 context samples, never touching the query). Table 5 specifies each of the 4 additional architecture variants' exact width/layer counts and parameter counts.

7. **Experiment Statistical Significance**
   - Question: Does the paper report error bars suitably and correctly defined, or other appropriate information about the statistical significance of the experiments?
   - Answer: [Yes]
   - Justification: Every headline, real-plant, ablation, and leave-one-family-out number is reported as mean$\pm$std over 5 independently-seeded training runs (Table 4 included: all five held-out-family variants reach $n = 5$ finite runs, guarded against the non-finite-gradient training divergence documented in the released repository's commit history). Section 4.1 explicitly discusses the limits of the ± 1 standard deviation heuristic used as a comparison bar (a descriptive, not multiplicity-corrected, statistic), and Section 5 separately flags a specific high-variance cell (Bouc-Wen `wh_only`, std 0.4163 on a mean of 1.9684) along with a documented single-seed-vs-5-seed reversal on that exact cell, rather than reporting only means.

8. **Experiments Compute Resource**
   - Question: For each experiment, does the paper provide sufficient information on the computer resources needed to reproduce the experiments?
   - Answer: [Yes]
   - Justification: All models were trained on a single NVIDIA GeForce RTX 2080 Ti (11GB) per run; no run requires multiple GPUs. Measured wall-clock for one 10,000-step training run of the default (1.58M-parameter) architecture on a dedicated GPU is approximately 14–16 minutes; larger ablation variants (up to 6.24M parameters) scale roughly with parameter count. The paper's trained-model results rest on approximately 50 total 10,000-step training runs (5 seeds $\times$ 2 recipes for the headline result; 4 architecture variants $\times$ 5 seeds for the ablation, with the default variant's 5 seeds reused from the headline runs; 4 held-out families $\times$ 5 seeds for the leave-one-family-out sweep, with the `backlash` family's 5 seeds likewise reused) — on the order of 12–13 GPU-hours total. Corpus generation itself is CPU-bound and separate from this budget: approximately 14 hours wall-clock on a heavily-shared, CPU-contended machine (see the released datasheet).

9. **Code Of Ethics**
   - Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics?
   - Answer: [Yes]
   - Justification: This work uses only synthetic, procedurally generated control-plant trajectories and publicly released real-plant system-identification benchmarks (Silverbox, Cascaded Tanks, Bouc-Wen), none of which involve human subjects, personally identifiable information, or scraped personal data. We are not aware of any way in which this research violates the NeurIPS Code of Ethics.

10. **Broader Impacts**
    - Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?
    - Answer: [Yes]

- Justification: Section 6 discusses positive impact (reducing duplicated private data-generation effort across the in-context SysID literature, and supporting excitation-design research via the released identifiability annotations) and states that we are not aware of misuse vectors specific to this release beyond those generic to open ML tooling.

11. **Safeguards**
    - Question: Does the paper describe safeguards for the responsible release of models with a high risk for misuse?
    - Answer: [NA]
    - Justification: The released models are small (1.58M–6.24M parameter) system-identification transformers trained exclusively on synthetic control-plant trajectories; they have no generative, language, or biological capability and no plausible high-misuse-risk profile of the kind this question targets. No additional safeguards beyond the license terms are applicable.
12. **Licenses**
    - Question: Are the creators or original owners of assets used in the paper properly credited, and are the license and terms of use explicitly mentioned and properly respected?
    - Answer: [Yes]
    - Justification: All existing assets used are cited with their creators and terms respected: Silverbox and Cascaded Tanks (nonlinearbenchmark.org, Section 4.2), Bouc-Wen [Noël and Schoukens, 2020] (CC BY-SA 4.0, DOI 10.4121/12967592, Section 4.2), the NeurIPS 2024 style template, and the datasheet template of Gebru et al. [2021] used for our own dataset's documentation.
13. **Assets**
    - Question: Are new assets introduced in the paper well documented, and is the documentation provided alongside the assets?
    - Answer: [Yes]
    - Justification: The new assets released by this paper — the PLANTFORGE corpus and all generation/training/evaluation code — are documented via a datasheet following the Gebru et al. [2021] template (motivation, composition, collection process, uses, distribution, and maintenance), a code license (MIT), and a data license (CC BY 4.0), released alongside the corpus and code (Section 3, and the repositories linked in Item 5 above).
14. **Crowdsourcing and Research with Human Subjects**
    - Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and details about compensation, if any?
    - Answer: [NA]
    - Justification: This paper involves no crowdsourcing and no human subjects research.
15. **IRB Approvals**
    - Question: Did the paper describe any potential participant risks and obtain Institutional Review Board approvals, if applicable?
    - Answer: [NA]
    - Justification: No human subjects were involved; no IRB review was required.
16. **Declaration of LLM usage**
    - Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research?
    - Answer: [NA]
    - Justification: Large language models are not a methodological component of the corpus generator, the excitation/identifiability code, or the trained in-context system-identification transformers themselves — all of which are ordinary procedural/simulation code and standard transformer training, with no LLM in the loop at

generation or inference time. In the interest of transparency beyond what this question strictly requires: an LLM-based coding assistant was used throughout this project's implementation, experiment orchestration, and paper drafting, under the sole author's direction and review, with every reported number independently re-verified against live code execution before inclusion in this paper.